

Probability with Combinatorics

Date_____ Period____

Find the probability of each event.

- 1) Cody is carrying seven pages of math homework and three pages of English homework. A gust of wind blows the pages out of his hands and he is only able to recover seven random pages. What is the probability that he recovers all of his math homework?
- 2) There are eight songs on your playlist. With random shuffle and no repetition, you listen to two songs. What is the probability that you listened to your favorite song first and your least favorite song second?
- 3) A test consists of ten true/false questions. A student who forgot to study guesses randomly on every question. What is the probability that the student answers exactly five questions correctly?
- 4) One day, eight babies are born at a hospital. Assuming each baby has an equal chance of being a boy or girl, what is the probability that exactly six of the eight babies are girls?
- 5) A fair coin is flipped seven times. What is the probability of the coin landing tails up at most five times?
- 6) A basketball player has a 50% chance of making each free throw. What is the probability that the player makes at most nine out of eleven free throws?